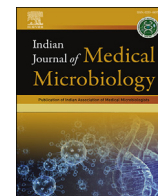




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Original Research Article

Genotypic characterization of carbapenem resistant Enterobacterales in clinical isolates from western Maharashtra

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ABSTRACT

Purpose: To study distribution of carbapenemase genes namely; New Delhi metallo-beta-lactamase (*bla*NDM), Oxacillinase-48 (*bla*OXA48), Verona Integron-Encoded Metallo-beta-lactamase (*bla*VIM) and Imipenemase (*bla*IMP) in carbapenem resistant *Enterobacterales* (CRE), isolated from clinical samples.

Method: This cross-sectional study was conducted at a tertiary care hospital of western Maharashtra over six months period. CREs were identified by conventional disc diffusion and modified carbapenem inactivation method (mCIM). A total of 50 consecutive CRE isolates from clinical samples were subjected to home brewed polymerase chain reaction (PCR) for detection of carbapenemases.

Results: Out of the 50 CRE isolates, at least one of the four carbapenemase genes was detected in 49 (98%) isolates. The frequency of distribution of these genes were NDM 90% (n = 45), OXA48 60% (n = 30) and VIM 12% (n = 6). Dual combination of *bla*NDM and *bla*OXA48 (50%) was the commonest pattern observed, which was frequently associated with *Klebsiella pneumoniae*.

Conclusions: The study indicate high prevalence of NDM warranting strict anti-microbial stewardship practices. Surveillance of CRE and resistance mechanism is essential to monitor the trend and take informed decision for appropriate anti-microbial therapy.

1. Introduction

Even almost a century after the discovery of first antibiotic we are still fighting against the microorganisms with even more desperation, trying to win the battle against the tiniest of the creatures. WHO have announced anti-microbial resistance (AMR) as an urgent priority area with several world leaders from world framing and executing their 'action plans' for the containment of AMR [1].

Enterobacteriaceae are a family of Gram-negative bacilli which falls within the domain Bacteria, phylum Proteobacteria, class Gammaproteobacteria, and order Enterobacterales [2]. They inhabit a wide variety of niches, including the human gastrointestinal tract, the gastrointestinal tract of other animals, and various environmental sites. They cause widespread infections ranging from self-limiting to life threatening infections and they are among the most common family of microorganisms responsible for causing wide variety of infections in humans [3].

Carbapenems namely imipenem and meropenem are beta-lactam group of antibiotics generally reserved for life-threatening multi-drug resistant Gram-negative infections. Carbapenem-resistant Enterobacterales (CRE) are posing alarming problem in the line of treatment,

with some showing resistance to all available antibiotics [4]. These infections increase length of stay and mortality rate in health care settings. Resistance to carbapenems is mainly mediated through the production of enzymes that inactivate carbapenems known as carbapenemases [5]. These enzymes have been divided into many classes (Ambler classification). Of these enzymes, the *Klebsiella pneumoniae* carbapenemase (KPC), first identified in 2001 in the United States, and New Delhi metallo beta-lactamase-1 (NDM-1), are the commonest [6]. Originally described in a *Klebsiella pneumoniae* isolate from India in 2008, NDM-1 enzymes have also been reported within the US, UK, and variety of other countries. These enzymes impart resistance to all or any beta-lactam antibiotics, except aztreonam [7]. The genes responsible for these enzymes reside on a mobile gene cassettes inserted into integrons that harbor additional antibiotic-resistance genes to other antimicrobial classes. As a result, this multidrug resistance (MDR) is often transferred to other bacteria by transposons and plasmids, severely limiting therapeutic options in serious infections [8].

In this era of globalization, CRE has spread rapidly worldwide due to extensive movement of humans. It is important to identify the frequency and types of carbapenemase prevalent in our country. This knowledge is

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not only for epidemiological importance but will also assist clinicians in instituting appropriate antimicrobial therapy. Therefore, our primary objective was to investigate distribution of carbapenemase genes namely; New Delhi metallo-beta-lactamase (*bla*NDM), Oxacillinase-48 (*bla*OXA48), Verona Integron-Encoded Metallo-beta-lactamase (*bla*VIM) and Imipenemase (*bla*IMP) in clinical isolates of CRE.

2. Materials and methods

This cross-sectional study was conducted at a tertiary care hospital of western Maharashtra over six months period. The study was approved by Institutional Review Board and Ethics Committee (Reference No: 22/AFMC/EC/2018 dated Oct 22, 2018). Clinical specimens received in microbiology laboratory were processed by automated as well as conventional technique by growth on suitable culture media. Isolate identification was performed by Gram stain, followed by conventional biochemical reaction or using VITEK-2 (Bio-merieux) automated bacterial identification system. Antibiotic sensitivity testing was performed by Kirby-Bauer disc diffusion method. CRE were identified by modified carbapenem inactivation method (mCIM) and the results were interpreted as per CLSI guidelines M100 2019 edition [9]. Genotypic characterization of carbapenemase was performed by home brew PCR. Briefly DNA extraction from pure colonies on nutrient agar was performed by AccuPrep Genomic DNA Extraction Kit, Bioneer Corporation. Forward and reverse primers for PCR amplification of genes encoding *bla*NDM, *bla*OXA48, *bla*VIM and *bla*IMP selected from prior published studies [10, 11] is shown in Table 1. A PCR master mix was prepared using Dream Taq Green PCR Master Mix (Thermo Scientific) 12.5 µl, 2.5 µl each of forward/reverse primer and 2.5 µl of nuclease free water. Extracted DNA template of 5 µl was added to 20 µl master-mix and subjected to thermocycling conditions as shown in Table 2. The data were tabulated in Microsoft excel sheet for analysis.

3. Results

A total of 50 non-repeat *Enterobacteriaceae* isolates found resistant to meropenem by both Kirby-Bauer disc diffusion and mCIM were selected for genotypic characterization. The distribution of clinical samples from which CRE were isolated is shown in (Fig. 1).

The mean age of our study population was 53 (range; 09–88) years, of these 42 (84%) were male.

Genotypic characterization by PCR showed presence of at least one of the four carbapenemases in 49 (98%) isolates (Fig. 2). Altogether,

Table 1

Primers used in the study for amplification of carbapenemases [10,11]. (F- forward, R - reverse, bp - base pairs).

Target genes	Primer pairs (3' - 5')	Amplicon size
<i>bla</i> NDM	F - CCGTATGAGTGATTGCGGCG R - GCCCAATATTATGCACCCGG	984 bp
<i>bla</i> OXA-48	F - GCTTGATCGCCCTCGATT R - GATTGCTCCGTGCGCGAAA	800 bp
<i>bla</i> VIM	F - GATGGTGTTTGGTCGCATA R - CGAATGCGCAGCACACAG	390 bp
<i>bla</i> IMP-2	F - GGAATAGAGTGGCTTAAYTCTC R - GGTTTAAYAAAACAACCACC	232 bp

Table 2

Polymerase chain reaction (PCR) protocol for amplification of carbapenemases.

Number of cycles	Temperature (°C)	Time	Purpose
1	95	5 min	Initial denaturation
35	95	45 s	Denatures DNA
	55	45 s	Anneals primers
	72	30 s	Extends DNA
1	72	8 min	Final extension
1	4	hold	Sample hold

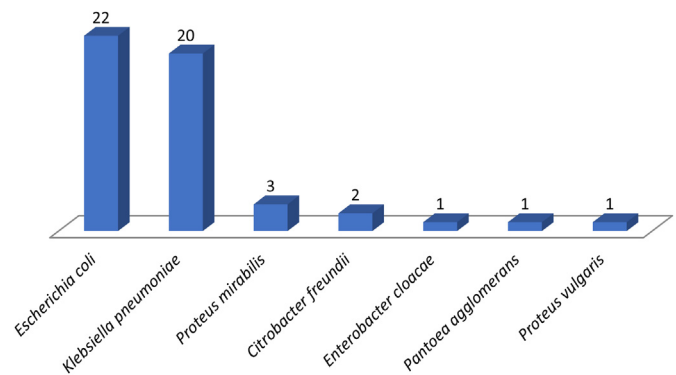


Fig. 1. Isolate distribution showing total number of isolates (N) on top of each bar of each species out of the 50 isolates.

*bla*NDM was most common carbapenemase, seen in 45 (90%) isolates. The frequency of other genes in decreasing order were, *bla*OXA48 60% (n = 30) and *bla*VIM 12% (n = 6). None of our isolate showed presence of *bla*IMP. Dual combination of *bla*NDM and *bla*OXA48 (50%) was the most common pattern observed, which was frequently associated with *Klebsiella pneumoniae*. (Table 3).

4. Discussion

CRE is dangerously emerging and posing a serious threat to health and its infrastructure limiting usage of available antibiotics for the treatment. It is important to detect them early so that preventive measures can be instituted early to curb its spread. Early identification of CRE is also important targeted antibiotic therapy.

Blood samples were commonest source of CRE isolation in our study as the study was conducted in a tertiary care hospital where septicemic patients with prior multiple antibiotic exposure gets frequently referred. The two most common CRE in the present study were *Escherichia coli* 22 (44%) and *Klebsiella pneumoniae* 20 (40%). Although the prevalence varies, *Escherichia coli* and *Klebsiella pneumoniae* were leading CRE in health care setting reported by other studies also [12–14].

Out of 50 isolates subjected to genotypic characterization, 49 isolates were PCR positive for at least one of the four genetic determinants of carbapenemases. Only 01 isolate (*Escherichia coli*) did not show any of the four carbapenemase on PCR, which indicate presence of other uncommon carbapenemase genes like *SME*, *IMI*, *PER*, *GES*, *SPM*, *SIM* etc [15]. Also other mechanism of carbapenem resistance may have been in this isolate. In isolation *bla*NDM 90% (n = 45) and *bla*OXA48 60% (n = 30) was by far the most common genes detected. Mariana Castanheira et al., also demonstrated that *bla*NDM gene is most common carbapenemase gene, followed by *bla*OXA48 gene [16].

NDM was first described in year 2009 and within just over a decade it has emerged as most common genetic mechanism for CRE. Present study showed NDM prevalence of 90% in CRE. The rapid spread is explained by plasmid mediated horizontal spread in bacteria of same species or other species. The emergence is also complemented by poor infection control practices. NDM is a metallo-β-lactamase enzyme which has zinc at its active site and therefore inhibited by metal ion chelators like EDTA. Organism with NDM is often multi drug resistant harboring other genes like extended spectrum beta lactamase (ESBL), other carbapenemase genes (as also evident by our study), cephalosporinases, 16S rDNA methylases etc rendering them resistant to carbapenems, other β-lactam antibiotics (except aztreonam), fluoroquinolones & aminoglycosides, therefore severely limiting the choice of antibiotics available for treatment of CRE organisms [17]. Studies have shown success rate of treating CRE bacteremic patients with drug like ceftazidime-avibactam (CAZ/AVI) and aztreonam-avibactam (ATM/AVI) to be 70.68% and 100.00% respectively [18].

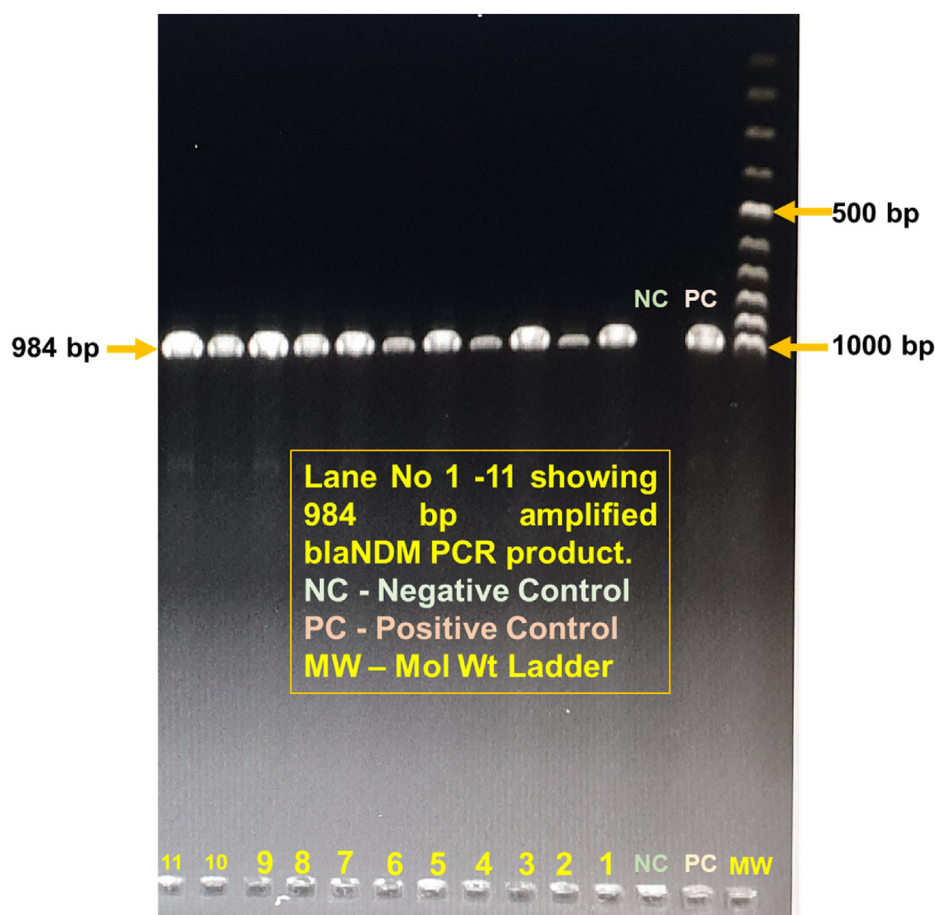


Fig. 2. Gel electrophoresis showing result of amplified PCR products of *blaNDM* (Molecular weight: 984 base pairs) along with positive control, negative control and molecular ladder (100 base pairs).

Table 3

Organism wise carbapenemase gene distributions.

Organism/Carbapenemase gene	<i>E. coli</i> (N = 22)	<i>K. pneumoniae</i> (N = 20)	<i>P. mirabilis</i> (N = 03)	<i>C. freundii</i> (N = 02)	Others (N = 03)	Total (N = 50)
NDM	06	05	01	–	02	14
OXA48	02	01	01	–	–	04
NDM + OXA48	10	14	–	01	–	25
NDM + VIM	02	–	–	01	01	04
NDM + OXA48 + VIM	01	–	–	–	–	01
VIM	–	–	01	–	–	01
Absent Carbapenemase	01	–	–	–	–	01
Total	22	20	03	02	03	50

Co-existence of NDM + OXA48 was found to be the most common pattern seen in 50% of the isolates. The isolates showing this pattern were *K. pneumoniae* ($n = 14$), *E. coli* ($n = 10$) and *C. freundii* ($n = 01$) respectively. Emergence and spread of such combinations further reduces therapeutic options. Co-existence of NDM + VIM was detected in 04 isolates (02 isolates of *E. coli*, 01 isolate of *P. vulgaris* and 01 isolate of *C. freundii*). Combination of NDM + OXA48 + VIM was seen in single isolate of *C. freundii*. India is endemic for NDM and the study shows various combinations of NDM with OXA48 and VIM. Lianyan Xie et al., showed co-existence of genes NDM and OXA48 in *K. pneumoniae* in their study [19]. Claudia Flores et al., showed various combinations of NDM producing *K. pneumoniae* with KPC, OXA48 and VIM [20].

The findings of the study were limited by small sample size and limited duration which may not reflect the idle burden of CRE in health care settings. Also, follow-up of CRE harboring individuals would have highlighted the outcome. Nevertheless, this study elucidates the mechanism of carbapenem resistance in a health care setting.

5. Conclusion

Emergence and spread of CRE is posing serious threat to human life and burden to health infrastructure. It is important to know the common mechanism and frequency of their resistance for epidemiological studies, diagnosis, as well as for choosing targeted therapeutic options. Widespread existence of CRE especially *blaNDM* producing CRE is a cause for concern, warranting strict isolation practices in health care setting. Surveillance of CRE and implementation of antibiotic stewardship program is the need of the hour.

CRediT authorship contribution statement

Suraj Giri: Conceptualization, Methodology, Writing – original draft, Investigation. **Santosh Karade:** Methodology, Supervision, Writing – review & editing. **Sourav Sen:** Visualization, Supervision, Validation, Writing – review & editing.

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